AMLGaurd: Graph Based Money Laundering Detection in Financial Networks

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***Abstract*—Detecting money laundering within financial networks is challenging due to the complexity of illicit transactions and the scarcity of labeled data. In this study, we model accounts as nodes and transactions as edges to develop an unsupervised anomaly detection framework, AMLGaurd, utilizing Graph Auto-Encoders (GAEs). GAEs encode the structural and transactional information of financial entities into a latent space and reconstruct the network to identify anomalies based on reconstruction discrepancies. We classify anomalies into four categories: contextual, structural, joint-type, and structure-type, each capturing different irregular patterns in the network. Furthermore, GAE encoder based embeddings can also be used for supervised edge classification, which can promote a multi-task learning setting. Although our present findings offer encouraging insights, this methodology establishes the groundwork for future progress in identifying complex money laundering schemes. This research enhances the application of graph-based unsupervised learning techniques in anti-money laundering (AML) systems, contributing to more effective and scalable financial security measures.**

***Index Terms*—Anomaly Detection; Anti-Money Laundering; Fraud Detection; Unsupervised Learning; Graph Neural Networks**

I. Introduction

THE detection and prevention of money laundering operations pose a significant challenge for financial industries, owing to the prevalence of annotated datasets and the complexity of illicit networks involved [1]. The objective of this research initiative is to apply unsupervised techniques for graph anomaly detection as part of the anti-money laundering (AML) system, leveraging the capabilities of Graph Auto-Encoders (GAEs) to uncover latent representations of financial transactions and relationships. We prioritize an unsupervised approach since it is extremely difficult to get real-world anti-money laundering (AML) datasets which do not often contain pre-labeled instances of fraud or money laundering. Also, traditional rule-based and supervised machine learning approaches often fall short in detecting complex money laundering patterns due to their reliability on predefined rule sets and labeled data.

Graph-based representations of financial networks aggregate a vast amount of transactional and relational data in a single structure [2]-[4]. Nonetheless, discerning and deriving patterns that matter from this data is complex, not least due to the nature of these networks. In this paper, we show that in homogeneous graphs (in which all nodes and edges are of the same type), Graph Auto-Encoders (GAEs) are an effective solution to this problem [5], [6]. GAEs are potentially powerful for latent representation as they encode the graph structure into a latent space and reconstruct it again, helping to discover the latent properties of nodes and their relationships [7], [8]. These can help unify critical node attribute and edge dynamics, leading to a deep understanding of underlying network structure. This knowledge is essential for accomplishing tasks such as node classification or edge classification that may be addressed under an unsupervised learning paradigm, such as without having any pre-labeled data. While these techniques effectively marginalize anomalous instances during encoding, existing proximity-driven reconstruction approaches exhibit critical limitations in anomaly detection. Current GAE methodologies [9], [10] effectively identify structurally clustered anomalies; however, they encounter difficulties with joint-type anomalies that do not aggregate naturally. The emerging research trajectory advocates for more holistic, neighborhood-centric strategies capable of capturing nuanced, non-locally clustered anomalous patterns within complex graph representations, thereby advancing unsupervised learning techniques.

Based on the above findings, we consider anomalies significant for AML efforts into four types: contextual, structural, joint-type and structural-type as represented in Fig. 1. **Contextual** anomalies are detected using substantial deviations of individual node features from anomalous behavior [11], [12]. **Structural** outliers exist as outlying subgraphs in the network structure (connected subgraphs with a very different structure), demonstrating unusual interaction behavior [13],[14]. **Joint-type** anomalies are characterized by the associating nodes having distinct output patterns and features, which indicates abnormal connections between segments [15],[16]. One of the more general categories of anomalies is **structure-type** anomalies, which can consist of structural and joint-type anomalies that refer to irregularities in the overall structure and connectivity of the network, respectively. We propose our novel approach that leverages Graph Auto-Encoders (GAEs) to better identify such anomalies with a forced reconstruction of the relevant neighborhood of a node. This refers not only to direct structural relationships, but also to the characteristics of

the node itself and its neighbors, based on the representation of an individual node. The methodology, in this case, aims at accurately identifying anomalies through a comparative analysis that makes use of the neighborhood reconstruction loss that is observed between abnormal nodes and their normal peers.

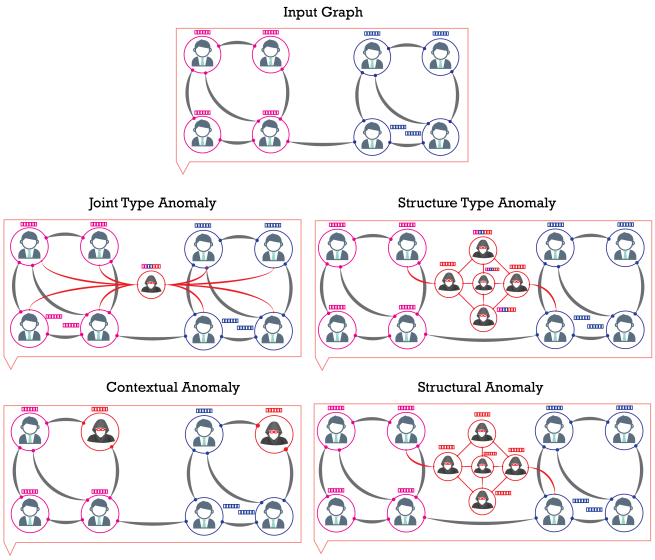

Fig. 1. **Diverse Categories of Anomalies.**

This work is in the domain of unsupervised learning with relevance for anti-money laundering and includes the application of Graph AutoEncoders (GAEs) to the anomaly detection task in financial networks. This paper is very much in the early stages and the goal of this study is to build a scalable framework that exploits latent feature learning of GAEs in the hopes of aiding in the detection of sophisticated money laundering schemes and improving the global financial security. As it explores this continuously, it aims to provide substance that can serve as the basis for the field. In summary, this study makes contributions as follows:

- We developed a novel framework AMLGaurd for graph based money laundering detection. We adopted neighborhood reconstruction approach to identify all variates of anomalies.
- Our proposed framework can be further used for multi-task classification purpose via extracting embeddings from the latent representation. This will enhance transaction level fraud detection.
- We constructed graph using IBM-AML dataset and conducted the experiment comparing with other real-world datasets.

II. Background

A. Overview of Money Laundering

Money laundering is the process of making illicitly acquired money appear lawful and introducing it into the normal financial system [17]. It is a serious problem that allows organized crime, drug trafficking, corruption and terrorism financing to prosper [18]. Money laundering has three main phases: placement, layering, and integration [19]. In placement phase, the illicit money is "processed" into the banking system, commonly by cash locations or acquisition of high worth assets. Layering phase is when convoluted transactions and movement happen to muddle the trail of cash. Finally, during the integration phase, the cleansed money is then reintroduced to the economy as apparently legit capital [20],[21].

B. Money Laundering Topologies (Patterns)

Money laundering typologies as shown in Fig. 2, refer to techniques criminals use to conceal the source of their illegally gained money [22]. **Fan-in**, when multiple smaller transactions are made that are then aggregated for a single larger transaction. The reverse happens in **fan-out**, where one big transaction is split into a number of smaller transactions to different accounts. A **cycle** is a type of transaction where one person sends money to another and, after a series of actions are carried out, they return to the person who transferred the money but there are some adjustments that make this a cycle and contributes to the cycle typology. In **bipartite** typology, the transactions involve two classes of entities: fund providers and fund recipients. **Stacked bipartite** is a closely related typology where there are several layers of intermediaries between the sender and receiver. This makes it hard to follow the source of funds by performing many transactions between multiple accounts in various global locations. With the **scatter-gather** typology, multiple small transactions are submitted to different accounts and aggregated before being moved to a single account. **Gather-scatter** refers to the input of funds to a single large transaction originating from multiple accounts, followed by a spread to several accounts making it difficult to trace the funds to the original source.

C. Related Work

Graph theory provides a powerful analytical framework in finance for detecting financial anomalies and identifying money laundering activities [23]. In transaction graph, nodes represent individual entities such as persons, corporations, or financial accounts, and edges represent fund transfers between these nodes [24]. Analyzing the structural properties and patterns within transaction graphs, suspicious activities like abnormal transaction volumes, unexpected entity connections, and sophisticated fund flows can be detected [25]. A very interesting direction here is the application of unsupervised graph neural networks (GNNs) yet such as those based on the Graph Autoencoder (GAE) architecture. GAE models discover low-dimensional representations (embeddings) of the graph nodes, retaining structural and descriptive information of the underlying financial network. The goal is to obtain an embedding of the graph that will be useful to identify anomalies as legitimate transactions and money laundering activities may lead to different patterns of behavior that can be seen in the structure of the graph and the properties of its nodes.

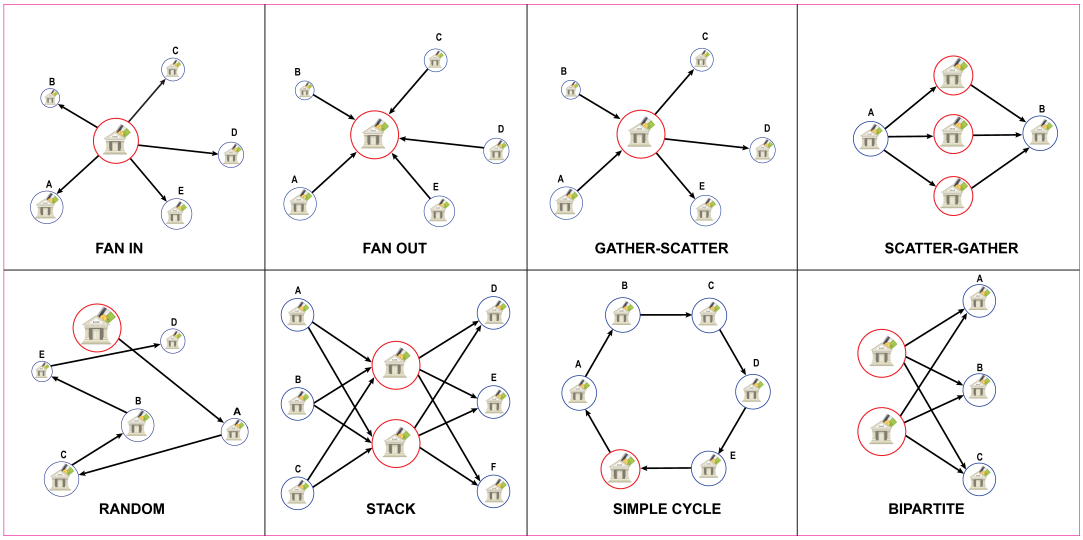

Fig. 2. **Money Laundering Patterns**

In the case of money laundering detection, the unsupervised nature of GAE models is extremely advantageous because real money laundering data is typically limited and hard to come by. As GAE models learn the financial network representation in an unsupervised way, they do not need labeled data in order to discover anomalous patterns. There have been multiple efforts applied unsupervised GNN-based approaches like GAE to money laundering detection. Fang et al. [27] proposes a multi-view graph neural network that utilizes both structure and attribute information to detect money laundering behaviors. Deng et al. [28] introduced a graph-based framework that combines transaction analysis and entity-level anomaly detection to enhance the identification of money laundering schemes. Unsupervised GNN models based on the GAE framework may also allow us to detect money laundering in a manner that is free of labeled data, but still informative enough for real world criminal activity investigation and prevention [26]. FlowScope [28] proposes a new metric, Anomalousness, which computes the likelihood for money laundering in multi-step streaming transaction graphs of high density with an end goal of tracing the full money flow (payments) from sources to destinations. Starnini et al. [29] find the smurf pattern and identify an AML algorithm that analyses two typical subgraph structures to locate smurf-like subgraphs. To improve the fraud detection precision, FRAUDAR [30] provides a new measure of subgraph density of suspicion focusing on disguised and non-disguised frauds. EigenPulse [31] also adopts a singular spectrum-based flow graphs in conjunction with a sliding window to model them with a row-enhanced matrix and capture surges in density in an efficient manner. A streaming graph-based model, MonLAD [32], monitors residual statistical features of the input stream for its explanatory power. SpecGreedy [33], a dense graph detection model, is based on graph spectral properties and uses a greedy peeling strategy for AML-type group mining. Together, these approaches seek either to find particular or dense subgraphs that correspond to money laundering activity.

III. Model Architecture

We adopt a neighborhood reconstruction approach [34] designed to detect anomalies in graph-structured data by integrating Graph Neural Networks (GNNs) with reconstruction-based techniques. The framework we propose leverages both unsupervised and supervised methods — an unsupervised node classification method for identifying money-laundering entities, and a supervised multi-task classification technique for edge classification that would highlight suspicious transactions. The architecture comprises two fundamental components — the encoder and the decoder.

Encoder: We utilize a GAE during the encoding phase that converts node features into dense embeddings. Every node collects messages from its neighbors and therefore contains information, both local and structural. Using a mix of feature transformation functions and aggregation operations, the encoder produces intermediate-level representations of the nodes that capture neighborhood and node-specific information. All the message-passing mechanism is thus very high extensible, covering single-hop to multi-hop message passing and different feature aggregation schemes like: Graph Convolution Networks (GCN) [36], GraphSAGE [37] Graph Attention Networks (GAT) [38] and Graph Isomorphism Networks (GIN) [39].

Decoder. In this part, we want to be able to reconstruct the features for each node as well as its neighborhood statistics to make it easier for us to do anomaly detection. In this, the decoder attempts to recreate the original node attributes and structural properties from the learned embeddings. The reconstructed values are then directly compared with the actual data using a weighted loss function that combines reconstruction errors for node features, node degrees and statistical

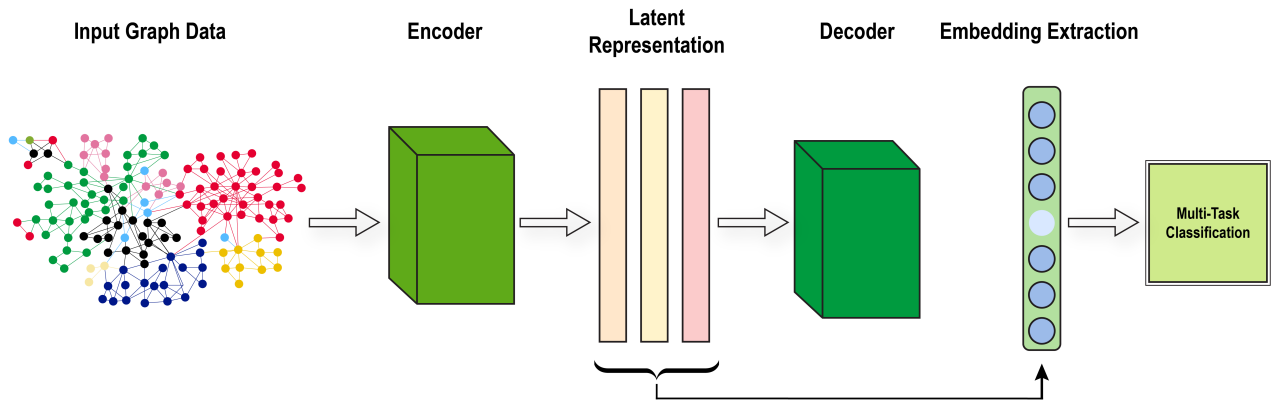

Fig. 3. **AMLGaurd Framework**

distributions of neighborhood representations. It incorporates multiple aspects of graph structure in a unified loss function that helps the model learn to identify deviations that resemble anomalies.

Algorithm 1 AMLGaurd: Graph Based Money Laundering Detection

Input: Graph $G(V, E)$, Input Feature X, Money Laundering Label Y

Encoder:

for *each* $u \in V$ **do**

  $h_u^{(0)} \leftarrow \xi(x_u)$

  $h_u^{(1)} \leftarrow \text{UPDATE}\left(h_u^{(0)}, \text{AGG}\left(\{h_v^{(0)} : v \in N_u\}\right)\right)$

  $H^{(1)} \leftarrow \{h_u^{(1)} : u \in V\}$ `// Node Embeddings`

  for *each edge* $(u, v) \in E$ **do**

    $h_{uv}^{(1)} \leftarrow \text{CONCAT}(h_u^{(1)}, h_v^{(1)})$ `// Concatenate node embeddings`

    $h_{uv}^{(1)} \leftarrow \text{MLP}(h_{uv}^{(1)})$ `// Transform edge embedding`

Decoder:

for *each* $u \in V$ **do**

  $\hat{h}_u^{(0)} \leftarrow \psi_x(h_u^{(1)})$, $\hat{d}_u \leftarrow \psi_d(h_u^{(1)})$

  $\mu_u \leftarrow \text{stop-gradient}\left(\frac{1}{d_u}\sum_{v \in N_u} h_v^{(0)}\right)$

  $\Sigma_u \leftarrow \text{stop-gradient}\left(\frac{1}{d_u - 1}\sum_{v \in N_u}(h_v^{(0)} - \mu_u)(h_v^{(0)} - \mu_u)^\top\right)$

  $\hat{\mu}_u \leftarrow \phi_\mu(h_u^{(1)})$, $\hat{\Sigma}_u \leftarrow \text{diag}\left(\exp(\phi_\sigma(h_u^{(1)}))\right)$

  for $i = 1$ **to** q **do**

    `// Reparameterization`

    $\bar{h}_i \leftarrow \text{FNN}(z_i)$, $z_i \sim N(\hat{\mu}_u, \hat{\Sigma}_u)$

  $\bar{\mu}_u \leftarrow \frac{1}{q}\sum_{i=1}^{q} \bar{h}_i$

  $\bar{\Sigma}_u \leftarrow \frac{1}{d_u - 1}\sum_{i=1}^{q}(\bar{h}_i - \bar{\mu}_u)(\bar{h}_i - \bar{\mu}_u)^\top$

  Weighted Loss Function:

  $L \leftarrow \lambda_x \sum_{u \in V} D(h_u^{(0)}, \hat{h}_u^{(0)}) + \lambda_d \sum_{u \in V} D(d_u, \hat{d}_u) + \lambda_n KL(N(\mu_u, \Sigma_u) || N(\bar{\mu}_u, \bar{\Sigma}_u))$

The model will classify the node as anomalous or normal based on the reconstruction loss. Nodes that produce large reconstruction errors are marked as potential money launderers. Furthermore, the encoder-generated embeddings are employed for multitask classification, wherein they are fed into edge classification tasks to identify fraudulent transactions. The framework provides a wide range of coverage by producing a dual-purpose approach for both account-level and transaction-level abnormalities in the overall financial network. Various types of anomalies are identified and detected using the proposed model based on the aforementioned encoder-decoder structure that reconstructs both the node attributes and the connectivity structure. Fig. 3 illustrates the proposed framework, which is also further described by the pseudocode in Algorithm 1.

IV. Dataset and Result Analysis

We constructed a financial graph network consisting of 515080 nodes and 5078345 edges using the IBM AML Small dataset [35]. Here, node represents the accounts, and edges are transactions between nodes. The initial dataset had transaction (edge) labels, from which node labels were generated; if a node had at least one illicit transaction (edge label), its node label was marked as 1 (illicit), otherwise 0 (legit) Initially, the graph had 1 node feature and 12 edge features.

TABLE I
Overview of Datasets Used

Dataset	# Nodes	# Edges	# Features
IBM AML Small	515,080	5078345	1
Disney	24	335	28
Books	1,418	3,695	21

In addition, we used two other datasets (Books [40] and Disney [41]) to compare the results. We try to learn the performance of each type of anomalies separately. We take the area under the ROC curve (AUC) as the evaluation metric. For the experiment, we treat the nodes of anomalies as positive classes and calculate curve for AUC. An area under the curve

(AUC) of 1 signifies perfect predictive accuracy, while an AUC of 0.5 denotes a lack of discriminative capacity in the model. Table 1. shows performance comparison (ROC-AUC) on IBM AML small dataset. We used GCN as message passing layer in encoder, and hyperparameters λ_x = 0.8, λ_d = 0.5 and λ_n = 0.001. By tuning the hyperparameters, we can give extra weight to specific types of anomaly detection.

TABLE II
PERFORMANCE COMPARISON (ROC-AUC) ON IBM AML SMALL DATASET

Hop	Structural	Contextual	Joint	Structure	Combined
1	67.32	55.05	51.59	50.24	65.06
2	54.19	53.24	51.66	53.95	55.22

The results presented in Table II show significant diversity of performance across both anomaly detection problems and GAE configurations. When considering GAE with a single GCN layer resulted to achieve much better AUC scores on structures and combined types of anomalies but however, having multiple layers in our models increased performance for structural-type anomalies types but did not influence contextual types. This means that simply making the network larger does not guarantee better discriminative performance on anomaly detection tasks. The accuracy of detecting normal and anomalous nodes was type-dependent, where joint-type anomalies were the hardest to detect correctly. In Table III, we have shown the results for two other datasets using GCN, SAGE, GAT and GIN as message passing layer with 2-hop feature aggregation. We kept the same hyperparameter settings during the experiment.

TABLE III
PERFORMANCE COMPARISON (ROC-AUC) ON OTHER DATASET

Dataset	Anomalies	GCN	SAGE	GAT	GIN
Books	Combined	65.34	65.83	44.95	64.79
	Contextual	62.34	56.48	57.02	58.37
	Structural	55.11	51.95	56.11	53.16
	Joint-type	48.77	49.90	58.54	48.10
	Structure-type	55.51	52.07	55.98	51.98
Disney	Combined	56.49	66.52	65.25	78.24
	Contextual	62.27	61.03	53.76	61.62
	Structural	56.05	58.09	57.72	54.73
	Joint-type	34.22	35.71	59.74	47.40
	Structure-type	55.07	57.91	57.91	54.34

V. CONCLUSION

In this study, we present a framework utilizing GAEs for financial anomaly detection. We have proposed a model for financial networks providing a classification of anomalies into four types that are designed to capture a range of irregular patterns within financial transactions. While the downstream multitask classification is still in progress, we encourage researchers to extend the work improving message passing algorithm as well as focus on dynamic edges to detect anomalies in real time.

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